

Raksha — Product Requirements Document

Product	Raksha — the real-time scam-call shield
Version	1.0 (hackathon MVP, production-shaped)
Authors	Shivam Gupta, with Claude
Event	YC Fall 2026 × Moss — Zero Latency Builder Sprint
Theme	Real-Time Voice & Conversational AI (with Local-First and Multiplayer elements)
YC RFS inspiration	<i>Proving You're Human</i> (Max Kolysh) · <i>AI for the Aging Population</i> (Max Kolysh)
Status	Built, tested, deployed

1. Problem

1.1 The one-line version

Phone scams are conversations with a script. Nobody protects people *during* the conversation.

1.2 Evidence

- Indians reported **₹22,845 crore** lost to cyber fraud in 2024 (up 206% year-on-year, ~10× 2022) across 22.68 lakh complaints; 2025 was flat at ₹22,495 crore. Cumulatively ≈ ₹53,000 crore over six years. (*MHA, Lok Sabha Q.432, Dec 2025*)
- Digital arrest** — fake police/CBI/customs officers who keep a victim on a video call and drain their savings — took **₹1,918 crore from 1,23,672 people in 2024** (+465% year-on-year); average loss per case ≈ ₹1.5–3.7 lakh. The Supreme Court took *suo motu* cognisance in October 2025. (*MHA data; SMW(CrI) 3/2025*)
- Losses concentrate on the elderly: single victims have lost ₹1.5 cr (a senior couple), ₹3.09 cr (Karnataka), ₹2 cr (a 92-year-old).
- Globally: FTC 2025 reported **\$15.9B** in fraud losses; phone calls have the **highest median loss of any channel (\$1,835)**, and **41% of \$10k+ losses by older adults began with a phone call**. FBI IC3 2025: victims aged 60+ lost **\$7.7B**, average > \$38,000. (*FTC, IC3*)
- The Arup case: one finance employee wired **US\$25.6M** after a video call with a deepfaked CFO and colleagues. (*CNN, 2024*)

Full sourced fact sheet: research/market-facts.md.

1.3 Why existing protection fails

Approach	Examples	Gap
Number reputation / spam labels	Truecaller (454M MAU), Airtel & Jio network AI, iOS 26 call screening, DoT Sanchar Saathi	Knows <i>who</i> is calling, not <i>what they are doing to you</i> . Crews rotate numbers; the script never changes.

Approach	Examples	Gap
Voice-liveness only	Truecaller AI Call Scanner	Detects synthetic voice, not social engineering by a human.
Content analysis, limited reach	Google Pixel Scam Detection (Pixel 9+, < 1% of Indian phones); Hiya AI Phone (cloud, US-only, \$9.99/mo)	Right idea, wrong distribution and wrong latency model for India.
Bank/telco signals	CBA–Telstra "Scam Indicator"	Knows <i>that</i> a call is happening at the moment of a transfer, not what was said.
Awareness campaigns	Mann Ki Baat "Ruko, Socho, Action Lo", caller tunes	Victims <i>know</i> the rules; the scam works by putting them in an emotional state where rules don't apply ("the ether").

Whitespace: multi-OEM, script-level detection with an *intervention* step (say this, alert a guardian, stop the transfer), fast enough to act inside one sentence.

1.4 Why now

- Token and embedding costs fall ~10×/year; on-device semantic search now runs in single-digit milliseconds (Moss).
- Regulation is moving liability onto banks: UK PSR mandatory reimbursement (Oct 2024, £354M reimbursed in 2025); **RBI's June 2026 directions make Indian banks compensate scam-induced losses from 1 Jan 2027** — the first time authorised-push-payment fraud carries a mandated cost for the bank.
- Deepfake voice makes "trust the voice" obsolete; the only durable signal left is the *script*.

2. Users

Persona	Situation	Job to be done
Sunita, 68, retired teacher, Bengaluru (protected person)	Gets a call from a "Mumbai Crime Branch officer". Alone at home, frightened, told not to tell anyone.	<i>"When someone is pressuring me, tell me clearly what is happening and what to say, before I do something I can't undo."</i>
Anita, 41, her daughter, Pune (guardian)	At work. Cannot answer every call; worries constantly.	<i>"Know the moment Mummy is in a dangerous call, and be able to reach her through the pressure."</i>
Fraud-prevention lead at a bank (buyer)	Owens reimbursement costs and 1930 escalations.	<i>"Stop the transfer before it happens, at a cost per customer far below the loss."</i>
Finance staff at a company (protected person, B2B)	"CFO" on a video call wants three urgent confidential wires.	<i>"Enforce the callback rule even when my boss is on the screen."</i>

3. Goals and non-goals

Goals (MVP)

1. Recognise a scam script from a live transcript **within one spoken sentence** of the first "ask", with retrieval in the **low tens of milliseconds** including on-device embedding (search itself sub-millisecond).

2. Intervene in a way a stressed, older person can act on: one headline, one sentence to say, one action, spoken aloud.
3. Never cry wolf on the calls that resemble scams most (a genuine bank fraud call, a courier, a hospital).
4. Break the isolation: a trusted person sees risk live and can speak into the call.
5. Learn from the network: a reported script protects every device within minutes, with no redeploy.
6. Be demonstrable by anyone in 60 seconds without a scam caller (scripted simulations with voices).

Non-goals (MVP)

- Deepfake audio detection (roadmap; complementary, not substitutable).
- Native telephony integration (Android call-audio capture, telco SIP hooks). **The MVP is a web app:** simulations and recordings work in any browser; the live-microphone mode needs a browser with the Web Speech API (Chrome, Edge, Safari) and a phone on speaker next to it. The phone app is the distribution vehicle and is the first roadmap item.
- Languages beyond English/Hinglish transcripts (playbook lines include Hinglish; STT language is configurable).
- Accounts, storage of calls, analytics.

4. Product

4.1 Surfaces

Surface	What it does
Shield (/shield)	The protected person's view. Modes: <i>Simulate</i> (9 scripted calls, two synthetic voices, real-time or fast, silent option), <i>Live mic</i> (Web Speech API, phone on speaker), <i>Recording</i> (upload → Whisper → replay). Risk dial, sparkline, transcript with tactic chips and per-line latency, playbook matches, coach card, full-screen intervention with spoken coaching, post-call summary and <i>Report this script</i> .
Guardian (/guardian?code=...)	Live dial and transcript highlights of a protected phone; interventions as they happen; <i>Speak through the shield</i> (shown and read aloud on the phone); <i>Ask the call</i> (semantic recall over the call's Moss session).
Playbook (/playbook)	Explore 31 scam families and 409 lines; live semantic search with engine/in-process/round-trip latencies; family detail with script arc and advice.
Latency lab (/lab)	Live benchmark against the running instance; the committed detection evaluation (detection rate, false alarms, family accuracy, time-to-danger, p50/p95/p99).
Landing (/)	Story, live runtime stats, how it works, why latency is the product, who pays.

4.2 The intervention ladder

Level	Trigger	What the person experiences
Caution (25–59)	First pressure tactics with confidence, or a coach "suspicious"	Amber toast: " <i>Careful — this call is starting to sound like a scam.</i> " + one sentence to say. Guardian notified.

Level	Trigger	What the person experiences
Danger (≥ 60)	The triad (pressure + ask), 4 distinct tactics, or coach "scam"	Full-screen red overlay, spoken aloud: script name, why, "Say this, then hang up", helpline. Simulation pauses. Guardian alerted.
Targeted ask (at danger)	A new ask tactic (OTP, transfer, remote access, personal info)	Louder, specific: "Do NOT read out the OTP."
Compliance (≥ 85)	The person begins to comply under pressure	"STOP. Put the phone down." — immediate, not rate-limited.

4.3 Functional requirements

ID	Requirement	Priority	Status
F1	Analyse interim STT fragments (≥ 4 words) every ≤ 500 ms and final segments on pause	P0	✓
F2	Retrieve top-K playbook lines per fragment with raw cosine scores, in-process	P0	✓ Moss query , alpha 1.0
F3	Deterministic risk state per call: tactics, family, stage, score, level, reasons	P0	✓ risk.ts , unit-tested
F4	Benign look-alike suppression using benign lines in the same index	P0	✓ 75 benign lines
F5	Speaker-aware crediting (person's words = compliance only)	P0	✓
F6	Intervention ladder with spoken output and rate limiting	P0	✓
F7	LLM coach on transitions with JSON contract and template fallback	P1	✓ Groq / any OpenAI-compatible
F8	Guardian rooms by family code; replay state on late join; speak-through; ask-the-call	P1	✓ Moss session query with call-id filter
F9	Community intel: opt-in report → upsert → hot-swap on all instances	P1	✓ raksha-intel , autoRefresh
F10	Nine scripted scenarios incl. two benign controls, playable with voices	P0	✓
F11	Live microphone mode (Chrome/Edge/Safari) with region-aware STT language	P1	✓
F12	Recording upload → Whisper → replay	P2	✓
F13	Health endpoint with runtime, LLM, latency stats	P1	✓
F14	Offline fallback retriever so the product runs without credentials	P1	✓ TF-IDF cosine
F15	Evaluation harness with committed report	P1	✓

4.4 Non-functional requirements

Area	Requirement	How it is met
Latency	Retrieval + scoring in the low tens of milliseconds p95 on a shared 4-vCPU container (search itself < 1 ms); intervention visible within one sentence	Moss in-process; risk engine O(tactics); interim fragments; measured in Latency lab and eval
Cost	Retrieval has zero marginal cost per fragment (local Moss queries are unmetered); the coach adds $\approx ₹0.02$ per call ($\leq \sim 6$ LLM calls on transitions)	Coach gated; free tiers sufficient for MVP

Area	Requirement	How it is met
Privacy	Server receives text only (browser STT); transcript in RAM only; recordings transcribed by Whisper and not stored; reports share caller lines only; no accounts	Architecture §8 and §4.7 below
Reliability	Runs with any subset of credentials; reconnecting WebSocket; call ends cleanly on disconnect; degrades to the offline retriever if Moss Cloud is unreachable and retries	Fallbacks + tests
Capacity	≥ 100 concurrent conversational calls per 4-vCPU container with zero errors	Measured: 100 calls, p50 12.8 ms / p95 58.0 ms, 0 errors (<code>npm run eval:load</code>)
Accessibility	Large type, high contrast, spoken output, single-action buttons	Design system
Portability	One container, any Docker host; Node 22	<code>Dockerfile</code>

4.5 Onboarding flow (no accounts)

1. The protected person opens the shield once; a six-character **family code** is generated and stored on that device only.
2. They tap *Copy link* and send it to a trusted person on WhatsApp, or read the code out.
3. The guardian opens the link; their browser joins the code's room and stays subscribed. Late joiners receive a replay of the current call state.
4. Every later call on the protected device broadcasts to that room automatically. Either side can rotate the code at any time; there is nothing to revoke because there is nothing stored.

4.6 How the score is computed

The 0–100 score is a **noisy-OR over persuasion tactics**: each retrieval hit's cosine similarity (from Moss, alpha 1.0) is calibrated to a 0–1 confidence, credited to the tactics tagged on that playbook line (after benign-look-alike suppression and speaker checks), and the per-tactic maximum is kept for the whole call. Two rules override the arithmetic: *pressure + ask* ⇒ at least 60 (Danger), and *the person begins to comply under pressure* ⇒ at least 85. Full definition, weights and unit tests:

[ARCHITECTURE.md §4](#).

4.7 Data handling

Data	Where it goes	Retention
Live audio	Stays in the browser's speech engine (Chrome/Edge: vendor speech service; Safari: on-device). Never sent to Raksha.	Not retained by Raksha
Uploaded recordings	Streamed to Whisper on Groq for transcription over TLS	Not stored by Raksha; Groq does not retain audio
Transcript text	Raksha server RAM, per call; the process's Moss <i>session</i> (local, in-process, turns tagged by call id) for recall	Turns deleted from the session at call end; record discarded after a 10-minute grace for the guardian summary
Coach prompts	Recent transcript text + risk summary to the LLM provider over TLS, only on risk transitions	Not stored by Raksha
Community reports	Caller's flagged lines only, opt-in, into the <code>raksha-intel</code> Moss index	Retained (shared knowledge); no personal data of the protected person

Data	Where it goes	Retention
Identity	None. No accounts, cookies or analytics; family codes are random capability tokens	—

Full privacy and threat model: [PRIVACY.md](#).

5. Success metrics

Metric	Target (MVP)	Measured
Scam scenarios reaching DANGER	100% of 10	100% (eval/REPORT.md)
Genuine-call scenarios reaching DANGER	0% of 8	0%
Everyday sentences (not in the index) credited with any tactic	< 5%	1.2% (1/80)
Turns from first "ask" to DANGER	≤ 1	see report
Retrieval p95 (embed + search)	< 30 ms on a shared container	14.9 ms (see report / Latency lab)
Demo-ability	Any judge can watch a scam stopped in ≤ 60 s with no setup	/shield?scenario=digital-arrest

North-star metric for the product: **₹ of transfers prevented per 1,000 protected calls**, measured by bank partners.

6. Business model and unit economics

Who pays

- Banks and payment apps (B2B2C, primary).** From 1 Jan 2027 RBI makes banks compensate scam-induced losses; UK banks already reimburse ~61% of APP fraud. LexisNexis puts the all-in cost of \$1 of fraud at > \$5. A per-protected-customer licence (₹5–15/month) that stops even a fraction of transfers is cheaper than any refund, and the bank gets a "transfer hold" signal it can act on in real time (the CBA–Telstra pattern, but with content).
- Telcos and OEMs.** A "protected line" add-on (₹49–99/month) on any Android phone, not just a Pixel; DoT's kill-switch ambitions need exactly this signal.
- Families (B2C).** A ₹99/month circle that protects parents — the price band where family-safety apps already convert (Life360: 2.8M paying circles, ARPPC \$143/yr).

Unit cost per protected call (10-minute call, ~120 fragments):

Item	Cost
Moss local queries	₹0 (unmetered)
Moss index storage/egress	fractions of a paisa amortised
STT	on-device (free) in the app; Whisper for recordings only
LLM coach (≤ 6 calls × ~600 tokens on a 20B model)	≈ ₹0.02
Hosting	one small container serves thousands of concurrent calls

≈ ₹0.02 per protected call. Worked example for the coach: $\leq 6 \text{ calls} \times \sim 900 \text{ input} + \sim 150 \text{ output tokens} \approx 6,300 \text{ tokens}$; at gpt-oss-20b list pricing (≈ \$0.10 per million input, \$0.50 per million output tokens on Groq) that is ≈ \$0.00024 ≈ ₹0.02. A single prevented ₹1.5-lakh digital-arrest transfer pays for ~7 million protected calls.

Pricing tiers (proposed)

Tier	Who	Price	Includes
Free	Individuals	₹0	Shield with simulations, live mic, one guardian
Family circle	Families protecting parents	₹99 / month	Up to 5 protected phones, unlimited guardians, call summaries, priority coach
Protected line	Telco / OEM bundle	₹49–99 / subscriber / month (wholesale ₹15–30)	White-label shield, carrier caller-context, regional playbooks
Bank shield	Banks and payment apps	₹5–15 / protected customer / month	Transfer-hold webhook, fraud-desk console, reimbursement analytics

Go-to-market: start B2C in India with the family circle (the emotional wedge: *protect your parents*), use the resulting call intel to sign one bank pilot on the transfer-hold signal, then bundle through a telco.

7. Competitive positioning

"Truecaller tells you who is calling. Raksha tells you what they are doing to you — and what to say back."

Defensibility comes from (a) the tactic-tagged playbook and the benign counter-set, which improve with every report, (b) the intervention UX validated on real people under stress, and (c) distribution through banks who now carry the liability.

8. Risks and mitigations

Risk	Mitigation
False alarms erode trust	Benign look-alikes in the index; speaker-aware crediting; coach veto; targeted rather than generic warnings; eval with benign controls.
Scripts drift (new pretexts)	Semantic matching generalises across paraphrase; community intel adds variants within minutes; playbook is data, not code.
STT quality in noisy rooms / Indian languages	Interim fragments tolerate partial text; region-aware STT; Whisper fallback; Hinglish lines in the playbook. Roadmap: on-device Indic STT.
Adversarial scammers slow down or split the ask	Evidence persists across the whole call; the triad fires whenever pressure and ask co-occur, however far apart.
Privacy concerns from listening	On-device STT; text-only, RAM-only; opt-in reporting of caller lines only; no accounts.
Platform access to call audio	Android supports accessibility/call-screening APIs; telco integration via SIP for the B2B path; the web reference implementation proves the runtime.

9. Roadmap

Horizon	Milestone
Now (MVP)	Everything in §4, deployed.

Horizon	Milestone
+4 weeks	Android app with in-call audio capture; Hindi/Tamil/Telugu STT; playbook lines in 5 languages; bank "transfer hold" webhook.
+3 months	Hot path fully on-device with the Moss browser/mobile SDKs (custom authenticator), so no text leaves the phone at all; deepfake-voice liveness signal fused into the risk model; pilot with one bank and one senior-living network.
+6 months	Federated playbook: reports from partners feed a shared intel index with regional variants; regulator reporting (1930/NCRP) from the app in one tap.

10. Open questions

- What is the right default for *who* can add a guardian — the protected person, or a family member during setup?
- How should the bank-side signal be priced: per protected customer, or per prevented transfer?
- Should the intervention ever auto-disconnect the call (Truecaller's family-protection can hang up remotely)? Our current stance: never without a human — the guardian or the person — in the loop.

Appendix A — The playbook

409 lines across 31 families (17 India-specific, 8 global, 5 US, 2 UK/AU), each tagged with 1–4 of 21 tactics, a severity (1–5), a script stage and a region; 75 benign look-alikes; 26 "victim about to comply" lines. Source and methodology: [research/scam-corpus.md](#). Data file: [data/playbook.json](#).

Appendix B — The tactic taxonomy

Authority · Urgency · Fear · Secrecy · Isolation · Payment demand · OTP/PIN request · Remote access · Skip verification · Too good to be true · Reciprocity · Personal-info request · Social proof · Escalation · Keep you on the line · Video-call demand · Legal threat · Account compromised · Relationship pretext · Tech pretext · You are about to comply.

Grounded in Cialdini's principles of influence, Stajano & Wilson's *seven principles* (2011), the FTC's *Four Signs of a Scam*, and I4C/RBI advisories on digital arrest.